WestQuant Memory Discovery Lab: An Evidence-Aware Framework from Experimental Data to Quantum-Memory Simulation

David Vesterlund
WestQuant, Vesterlund Ventures, Sweden
Independent research program

10 October 2026

Abstract

Quantum-memory modeling requires measured observables, control conditions, optical transfer and acceptance to be distinguished. We present WestQuant Memory Discovery Lab, an evidence-aware framework linking versioned data to reviewed coherence references, supplied dynamics and scoped network instruments. Reanalysis of 90 released Pr^{3+}:Y_2SiO_5 fidelity values recovers path and time-bin means of 0.951082 and 0.914048; eight fits to separate retrieval curves give six mismatches and two inconclusive results. A distinct $^{13}C/NV^0$ dataset supplies 24 measured observables: a calibrated envelope passes a held-out diagnostic (maximum scaled residual 2.428), whereas signed Ramsey X fails (3.158, threshold three), preventing physical-model promotion. Joint write-store-read/detection instruments pass 56 channel-age and 336 state-probe comparisons against an independent numerical reference, with maximum discrepancy 1.443×10^{-15}. Six convergence controls comprise 69 coherence calculations on bounded synthetic baths up to eight spins. A native SeQUeNCe/QuNetSim comparison then completes 256 simulator runs and 131 072 trials; all 128 paired comparisons agree, with maximum mean-density discrepancy 2.465×10^{-15}. QuNetSim uses a protocol-specific QuTiP backend and an imposed logical clock, so physical network latency is not cross-validated. A fresh offline replay checks every native run and random trace; 107 tests pass and 2 optional tests are skipped in the current environment. Matched experimental fidelity/efficiency/noise data and validated new-material parameters remain missing. The results establish a reproducible integration and verification workflow, not a new material, improved hardware performance or universal simulator validation.

Keywords: quantum memories; reproducible simulation; provenance; open experimental data; spin coherence; quantum channels; quantum networks.

1 Introduction

The useful lifetime of a quantum memory depends on the task being performed. Spin coherence, optical coherence, storage efficiency and conditional qubit fidelity are different observables. A material may preserve a spin superposition under a specialized pulse sequence while lacking an efficient or sufficiently quiet optical write/read interface. Consequently, a numerical value labeled T_2 cannot by itself determine whether a device supplies useful states to a quantum network.

Published experiments illustrate these distinctions. Teller et al. demonstrated individually addressable memory cells and measured path- and time-bin-qubit fidelities with weak coherent probes [1]. Ortu et al. reported protocol-dependent effective spin-coherence times together with time-bin storage [3]. A recent Pr-memory experiment distinguishes extended classical echoes from nonclassical heralded-photon storage [4]. Molecular-qubit studies supply complementary

information about phase memory and chemical environments [6, 7], whereas integrated defect experiments report distinct Ramsey and Hahn observables [9]. These data support different model components and should not be pooled as interchangeable lifetime labels.

Existing computational tools already solve much of the relevant physics. PySCF provides electronic-structure calculations [10]; PyCCE evaluates spin-bath dynamics [11]; QuTiP supplies open-system evolution [12]; and MaxwellBloch treats optical propagation [13]. SeQUeNCe supplies event-driven network simulation [17], while QuNetSim provides a layered quantum-network framework [16]. General workflow systems such as AiiDA address computational provenance [18]. Recent cross-validation of network simulators emphasizes that matching parameter names is insufficient when event and connection semantics differ [19]. WestQuant reuses these tools. It does not replace their solvers, and the implementation evaluated here does not use AiiDA as its execution engine.

The research question is whether a common executable workflow can move from heterogeneous experimental and materials data to appropriately scoped memory simulations while retaining the evidence needed to assess each result. We address it with a typed catalog, condition-sensitive reference queries, explicit model dispatch, an acceptance-and-erasure channel contract and archived validation outputs. The same representation permits controlled candidate variations, but a generated candidate is not assumed to be a stable species or a functioning memory.

The contribution has three parts: an implemented source-to-model interface, a common memory/channel contract with declared inference boundaries, and reproducible evaluations using open experimental data and independent numerical references. Existing-memory validation supports the system contribution. Finding a better material, changing a network ranking or demonstrating a new microscopic prediction is not required for that contribution and is not claimed here. Machine learning for T_2 has been separated into another project and is absent from the evaluated workflow.

2 System architecture and evidence model

Figure 1 summarizes the implemented workflow. Each source artifact retains its URL, revision or record identifier, raw bytes, content hash, retrieval metadata and license information. Bounded downloads check source checksums when supplied. Offline execution checks the cached SHA-256 values and fails on missing or corrupted artifacts. A retrieval timestamp alone is not treated as an immutable source revision: live reference pages are frozen by their retrieved bytes and hashes.

2.1 Candidates, observations and references

A candidate record contains elemental composition, explicit charge, electronic spin, supplied geometry, host and any site-resolved isotope substitutions. Candidate enumeration checks electron-count/spin parity and preserves the parent configuration. These checks remove inconsistent specifications; they do not establish a bound electronic state, chemical stability, accessible optical transitions or an experimentally realizable device.

An observation contains a subject identifier, quantity, value, declared unit, conditions, source location, provenance and evidence kind. Distinct kinds include experimental, curated experimental, evaluated reference, computed, hypothetical and unknown evidence. Result metadata separately records input and output evidence. For example, a computed channel constrained by a published mean remains a modeled channel rather than an experimental process-tomography result.

The catalog uses SQLite with immutable record identities. The coherence-reference extension stores a definition, a positive time in seconds, reported uncertainty convention, explicit conditions, primary-source location and unresolved conditions. Reference matching requires material identity and coherence definition, then compares the supplied reported conditions. Missing, mismatched

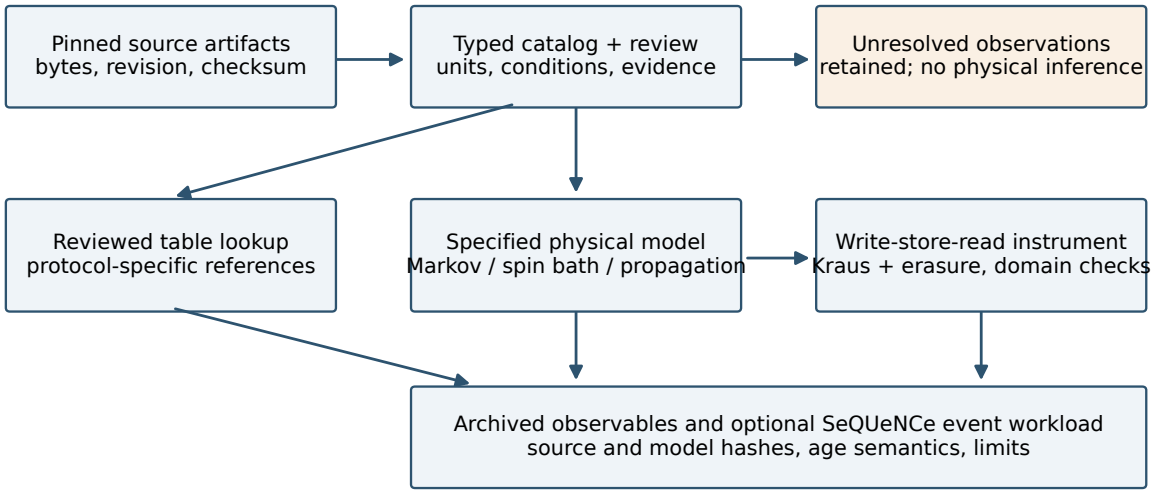


Figure 1: Implemented data and model boundaries. Unresolved observations remain visible in the catalog. Reviewed table values and specified physical models can produce scoped outputs; optical channel construction additionally requires write/read and loss parameters. SeQUeNCe is an optional downstream workload. A table reference is not automatically converted into a full channel.

or unverifiable conditions cannot silently yield an applicable numerical estimate. Even a reported-condition match retains the source's unresolved microscopic or operating details. No temperature, isotope, host, charge or pulse-protocol interpolation is performed by the reference-query path.

This restriction differs from the separate *pure-coherence table* model: a reviewed complex-coherence curve may be interpolated linearly within its own time interval under matching temperature and control conditions. Its initial value must be unity, its magnitude must remain within the unit disk, and it must explicitly exclude population relaxation. Extrapolation is prohibited. Neither table mechanism converts a bare formula or an isotope mass into T_2.

2.2 Execution and failure semantics

Study execution enumerates configuration identities before launching bounded worker processes. Hashes include configuration and source information; random streams are derived from explicit master seeds and identities. Atomic JSON writes, checkpoints and resume logic preserve completed records. Invalid geometry, unsupported model families, missing units, incompatible temperature domains and numerical nonconvergence are failures or unresolved outcomes rather than substituted defaults. A generic study-stage gate is a workflow control, not a sufficient proof of scientific discovery; the baseline's richer discovery gate must also be interpreted at its documented scope.

The implemented dispatcher groups candidates into atomic ensembles, trapped atoms/ions, rare-earth solids, defect spins, molecular spins and a restricted negative-ion branch. This is a software taxonomy with six labels, not an exhaustive classification of quantum memories. The allowed model combinations are listed in Table 1. Quantum dots and optomechanical devices do not have independently validated microscopic adapters in this release.

Family or component	Implemented scope
Atomic ensembles; trapped atoms/ions	Supplied Markov model or condition-matched pure-coherence table; no general first-principles device identification.
Rare-earth, defect and molecular spins	Above descriptions plus finite nuclear-bath evolution or conventional PyCCE coherence with supplied parameters.
Negative ion	Supplied Markov/table storage and a specified photodetachment environment-overlap model; no inferred bound state or coherent optical interface.
Electronic and propagation backends	Molecular PySCF; ORCA/QE input and coupling-import hooks; a separate MaxwellBloch propagation adapter. These do not automatically identify a memory channel.
Experimental operating point	A reviewed rare-earth point with exact time/protocol/temperature matching; lifetime extrapolation rejected.

Table 1: Implemented model domains. Availability of an adapter and numerical execution are separate from experimental validation of that domain.

3 Physical and operational model

3.1 Stage-resolved conditions and Markov dynamics

A modeled memory specifies writing, storage and reading temperatures (T_w, T_s, T_r) and durations (t_w, t, t_r). Each stage is isothermal in the evaluated implementation. An unspecified thermal ramp is rejected; a temperature-dependent coefficient is used only inside its declared validity interval. Phenomenological rate functions can be supplied as

$$\Gamma(T) = \Gamma_0 + aT + b(T/T_{\mathrm{ref}})^n + \frac{c}{\exp(\Theta/T) - 1}, \tag{1}$$

with nonnegative coefficients and an explicit reference temperature. These coefficients are inputs or calibrated model quantities. Equation (1) does not establish a material-specific relaxation mechanism merely because terms resemble direct, Raman or activated processes.

For stage j, the Markov reference uses population relaxation $\Gamma_{1,j}$, independent pure dephasing $\Gamma_{\phi,j}$, excited-state equilibrium population $p_{e,j}$ and detuning ν. With $H = h\nu|1\rangle\langle 1|$ and $\mathcal{L}_A(\rho) = A\rho A^\dagger - \{A^\dagger A, \rho\}/2$,

$$\dot{\rho} = -\frac{i}{\hbar}[H, \rho] + \Gamma_1(1 - p_e)\mathcal{L}_{\sigma_-}(\rho) + \Gamma_1 p_e \mathcal{L}_{\sigma_+}(\rho) + \frac{\Gamma_\phi}{2}\mathcal{L}_Z(\rho), \tag{2}$$

$$\rho_{01}(t) = \rho_{01}(0)\exp[-(\Gamma_1/2 + \Gamma_\phi)t + i2\pi\nu t]. \tag{3}$$

Thermal detailed balance is optional and, when declared, uses a supplied qubit frequency to determine p_e. A separately declared activity contribution can add a phenomenological rate λa to storage dephasing; it is not an independently validated model of network-induced heating or pulse errors.

Under this stationary, independent-noise description,

$$T_1 = \Gamma_1^{-1}, \qquad T_2 = (\Gamma_1/2 + \Gamma_\phi)^{-1}. \tag{4}$$

An experimentally reported *total* T_2 must not be inserted as $\Gamma_\phi = 1/T_2$ and combined with population relaxation again. The model requires an explicit rate definition to prevent this double counting. The familiar $T_2 \leq 2T_1$ statement is conditional on this model and is not used as a universal filter for protocol-dependent fitted times.

3.2 Finite nuclear-bath evolution

Hamiltonians in the spin module are frequency operators G in Hz; physical energies are hG. The supplied bath contains nuclear spins I_i, laboratory-frame Zeeman vectors $\boldsymbol{\omega}_i$, hyperfine tensors A_i,

optional symmetric quadrupole tensors Q_i and pair tensors J_{ij}:

$$G_B = \sum_i \boldsymbol{\omega}_i \cdot \mathbf{I}_i + \sum_i \sum_{ab} Q_{i,ab} I_{i,a} I_{i,b} + \sum_{i<j} \sum_{ab} J_{ij,ab} I_{i,a} I_{j,b}, \tag{5}$$

$$G_m = G_B + m \sum_i \sum_a A_{i,za} I_{i,a}. \tag{6}$$

The central-spin pure-dephasing approximation retains the central S_z coupling and the two specified logical projections m_0, m_1. Transverse central-spin transitions and phonon relaxation are not generated by this approximation. The original controls use Gibbs initialization,

$$\rho_B(T) = \frac{\exp[-hG_B/(k_B T)]}{\mathrm{Tr}\exp[-hG_B/(k_B T)]}, \qquad L(t) = \mathrm{Tr}[U_0(t)\rho_B(T)U_1^\dagger(t)]. \tag{7}$$

The extension also supports explicitly maximally mixed initialization, $\rho_B = I/d$, used for its PyCCE/exact convergence comparisons. The zero-temperature Gibbs implementation uses the normalized projector onto the numerically degenerate ground-state subspace. Free propagators contain $\exp(-i2\pi G_m \Delta t)$. Ideal CPMG pulses swap the two branch Hamiltonians at $t(j+1/2)/N$, $j = 0, \ldots, N-1$. A finite Hilbert-space budget bounds exact evolution.

Conventional CCE is available through PyCCE [11]. Its backend units are explicitly converted: seconds to milliseconds, hyperfine Hz to kHz and field tesla to gauss. Generalized CCE density elements are not silently normalized and exported as pure coherence. The extension explicitly passes supplied pair tensors to PyCCE and checks Zeeman conventions when required. Bath-size, cluster-order, pair-cutoff, propagation-resolution and time-grid convergence are examined for synthetic controls in Section 8; none validates a new material.

The operational descriptor returned by exact evolution is the first *sampled* $1/e$ crossing of $|L|$, represented by adjacent time-grid endpoints. Revival detection and the finite observation window accompany it. This descriptor is not declared to be a unique exponential T_2. Absence of a sampled crossing is not proof that none occurs between samples, and is not reported as an infinite measured lifetime.

3.3 Electronic structure, optical propagation and negative ions

The molecular electronic adapter evaluates HF or DFT at supplied geometry, charge and spin, optionally with scalar relativistic treatment and Tamm–Dancoff excitations. Convergence is checked and recipes are included in the cache identity. A finite-basis energy does not prove an anion bound state; a vertical excitation energy does not specify T_2. Couplings from external calculations require explicit site mapping, tensor-frame alignment and units before entering Eq. (6). The evaluated workflow does not automatically calculate a complete spin–phonon spectral density from a molecular structure.

The MaxwellBloch adapter evolves a supplied multi-level optical recipe separately from the qubit-channel wrapper. It can provide propagation fields and states, but a finite-field solution is not automatically a single-photon acceptance instrument. Protocol calibration and an observable mapping remain necessary.

For photodetachment, neutralization leaves an electron or environmental record. For a modeled superposition

$$|\Psi\rangle = a|0\rangle|E_0\rangle + b|1\rangle|E_1\rangle, \qquad \rho_{01} \longmapsto \chi\rho_{01}, \quad \chi = \langle E_1|E_0\rangle, \tag{8}$$

tracing out that record reduces coherence unless the records overlap sufficiently. The restricted negative-ion branch therefore requires a supplied complex χ with $|\chi| \leq 1$. It does not infer χ from a charge change or assert that photodetachment produces a stable quantum memory.

3.4 Write-store-read channel and erasure

Let Φ_j be the conditional stage channel from Markov dynamics, and let $\mathcal{D}_z$ multiply ρ_{01} by complex z, with $|z| \leq 1$. The storage map is

$$\Phi_s(t) = \mathcal{D}_{L(t)} \circ \Phi_{s,\mathrm{Markov}}(t), \qquad \Phi(t) = \Phi_r \circ \Phi_s(t) \circ \Phi_w. \tag{9}$$

Write/read depolarization is separately supplied. Composition with bath or table coherence is allowed only when its independence from the Markov contribution is explicitly declared. The optical acceptance parameters η_w, η_r and independent erasure rates yield

$$\eta(t) = \eta_w \eta_r \exp[-\Gamma_{e,w} t_w - \Gamma_{e,s} t - \Gamma_{e,r} t_r]. \tag{10}$$

This separable expression is a model assumption, not a universal relation between optical efficiency and spin coherence.

The implemented complete map sends a qubit to a qubit-plus-erasure output:

$$\mathcal{E}_t(\rho) = \eta(t)\Phi_t(\rho) \oplus (1 - \eta(t)) \operatorname{Tr}(\rho)|e\rangle\langle e|. \tag{11}$$

If $\{K_a\}$ is a Kraus representation of Φ_t, accepted operators are the 3×2 matrices obtained by embedding $\sqrt{\eta} K_a$ in the logical output block. Two loss operators are $\sqrt{1-\eta}|e\rangle\langle 0|$ and $\sqrt{1-\eta}|e\rangle\langle 1|$. Their adjoint products sum to the identity. The unnormalized, column-vectorized Choi matrix is

$$J(\mathcal{E}) = \sum_a \operatorname{vec}(K_a) \operatorname{vec}(K_a)^\dagger, \quad J \succeq 0, \quad \operatorname{Tr}_{\mathrm{out}} J = I_{\mathrm{in}}. \tag{12}$$

Kraus compression uses Choi eigendecomposition with a declared numerical cutoff. Complete positivity, trace preservation and state validity are checked numerically. These tests establish a physical map under its assumptions; they do not establish that the map is the correct device model.

Equation (11) assumes state-independent acceptance and ideal loss flags. State-dependent acceptance requires a general trace-decreasing quantum instrument; dividing its output by a state-dependent success probability would produce a nonlinear conditional transformation. That broader instrument is outside the implemented export contract.

3.5 Readout background and a complete effective instrument

The signal-only erasure map is extended with a named background source independent of the noise already in Φ_t. The event model accepts a retrieved signal first; background is accepted only if signal retrieval failed. If $r(t) = r_0 + at$ is the supplied background rate and Δ the detector window,

$$b(t) = 1 - e^{-r(t)\Delta}, \qquad q(t) = [1 - \eta(t)]b(t), \qquad p_{\mathrm{acc}}(t) = \eta(t) + q(t). \tag{13}$$

For a fixed background qubit state σ, the complete linear instrument is

$$\mathcal{D}_t(\rho) = [\eta(t)\Phi_t(\rho) + q(t)\sigma \operatorname{Tr}\rho] \oplus [1 - p_{\mathrm{acc}}(t)] \operatorname{Tr}\rho |e\rangle\langle e|. \tag{14}$$

Signal efficiency η, noise-click probability q and total acceptance p_{acc} remain separate. A fixed-state background has six-Pauli-state average target overlap $1/2$, giving

$$F_{\mathrm{det}}(t) = \frac{\eta(t) F_{\mathrm{signal}}(t) + q(t)/2}{p_{\mathrm{acc}}(t)}, \qquad O_{\mathrm{uncond}}(t) = p_{\mathrm{acc}}(t) F_{\mathrm{det}}(t). \tag{15}$$

Conditional fidelity is undefined when $p_{\mathrm{acc}} = 0$ and is stored as missing. Duplicate noise-source identities are rejected. This is an effective state-independent first-click instrument; it excludes signal/background double clicks, multiphoton physics and state-dependent detection, and is not itself an experimentally identified detector model.

3.6 Absolute age and network semantics

The exported memory device stores channels at explicitly requested ages. Retrieval applies the complete age-dependent channel once to the original input state; incremental application is rejected. This avoids applying the same decay twice and avoids a semigroup assumption in revival-containing dynamics. It supports fixed, declared control histories, not arbitrary interventions on a non-Markovian bath. The network adapter performs no age interpolation or extrapolation: an absent sample is a missing-data error.

The SeQUeNCe adapter acts on the selected key in its complete joint density matrix, with explicit tensor ordering. It samples loss before applying conditional channels, resets consumed or erased memory states and does not add SeQUeNCe's default memory decoherence a second time. Event durations are converted to integer picoseconds with a representability check. The evaluated workload loads an ideal Bell pair, waits for a fixed age and attempts retrieval from two nodes. It uses the actual event kernel and density-matrix manager, not a full routing, swapping or realistic photon-generation stack.

4 Observable definitions and inference gates

4.1 Fidelity and useful output

For an input $|\psi\rangle$, conditional target fidelity is $F_\psi = \langle\psi|\Phi(|\psi\rangle\langle\psi|)|\psi\rangle$. A uniform six-Pauli-state mean equals Haar-average qubit fidelity only for one linear channel and appropriate state-independent conditioning. For that model,

$$F_{\mathrm{av}} = \frac{2F_e + 1}{3}, \qquad F_e = \frac{1}{4}\sum_a |\operatorname{Tr} K_a|^2. \tag{16}$$

The framework separately reports F_{av}, success probability and unconditional target overlap ηF_{av}. These quantities should not be merged into one unspecified "fidelity." A classical benchmark for weak coherent probes also cannot be replaced by a universal 2/3 threshold without the relevant photon-statistics and efficiency model.

In the two-node workload, a chosen Bell target defines F_B after retrieval. For a specified threshold $F_{\min}$, the quality-qualified delivery rate is the number of accepted pairs with $F_B \geq F_{\min}$ divided by total simulated attempt time. The threshold is a benchmark parameter, not a QKD security guarantee. The implementation does not claim that this closed workload supplies a realistic end-to-end network rate.

4.2 Protocol-specific coherence inference

The data representation distinguishes Ramsey T_2^*, Hahn coherence, DD-dependent effective spin T_2, molecular phase-memory time T_m and effective optical AFC coherence. For example, an AFC-efficiency expression with exponent factor four and a spin-wave intensity model with exponent factor two cannot be interchanged [3]. A single fidelity point cannot determine a lifetime.

For curve diagnostics, the Gaussian likelihood uses supplied uncertainties with

$$y(t) = b + A\exp[-f(t/\tau)^\beta], \tag{17}$$

where $\beta = 1$ for an exponential and $0.25 \leq \beta \leq 4$ for the stretched model. The positive factor f must have an explicit observable mapping. In the Pr-efficiency diagnostics below, $f = 1$ defines an *apparent retrieval-decay scale*; it is not converted to physical spin T_2.

For supplied phase-resolved counts, $k_i \sim \mathrm{Binomial}(n_i, p_i)$ and

$$p_i = \tfrac{1}{2}\{1 + b + A\exp[-f(t_i/\tau)^\beta]\cos\phi_i\}. \tag{18}$$

Least-squares minimization of signed square-root binomial-deviance residuals implements the corresponding likelihood fit. The evaluated count control has known phases; phase drift is not fitted away as decay.

Fits use three starting scales and bounded log-time parameters. Every fourth unique time group is held out, including all phases at that time. Gaussian standardized residuals or binomial Pearson residuals give a held-out RMSE. Model mismatch is flagged when the approximate full-data goodness-of-fit diagnostic has $p < 0.01$ or held-out RMSE exceeds three. Missing mapping review, boundary estimates, a scale outside the sampled interval, or unstable/wide bootstrap intervals produce an inconclusive outcome. These are explicit screening diagnostics under the declared error model, not calibrated general-purpose hypothesis tests.

Parametric bootstrap intervals condition on the fitted model and supplied likelihood. They exclude shared systematics and model uncertainty. A negative diagnostic is retained; the software never selects an unreviewed numerical scale simply because a solver returned a value. In particular, an efficiency-decay fit and a correlation $g^{(2)}$ measurement do not supply qubit process tomography.

5 Evaluation design and source audit

All analyses are retrospective. Splits and gates are explicit in the configuration and code, but were not prospectively preregistered. We distinguish four forms of evidence: data-consistency/reanalysis checks, within-dataset model diagnostics, mathematical/software verification, and optional simulator integration. No new hardware experiment or independent experimental replication was performed.

The first experimental case uses Teller et al.'s released processed data [1, 2]. Fourteen source artifacts are cached and checksummed. Ninety state-fidelity scalars cover six probe states in five path pairs and ten time-bin cells. These scalars are not ninety independent experiments. Source indexing is checked against released summary values; inconsistent headers and an ambiguous time axis are documented rather than silently trusted.

The spatial holdout calibrates a common conditional-fidelity mean on cells 1, 3, 5, 7 and 9 and assesses cells 2, 4, 6, 8 and 10. Inverse-variance weighting uses the supplied Monte Carlo uncertainty widths. The diagnostic is

$$z_i = \frac{|F_i - \widehat{F}|}{\sqrt{\sigma_i^2 + \sigma_{\widehat{F}}^2}}. \tag{19}$$

Cross-cell covariance and common systematics are unavailable; the gate $\max_i z_i \leq 3$ therefore supports only an internal approximation check. It does not validate a unique channel or the separately calibrated path point.

A separate Pr-memory retrieval dataset uses the seven released measurement files in Zenodo record 21532904 [4, 5]. Four classical curves provide 15 no-RF and 70 observations per RF sequence (XX, XY4, XY16), totaling 225 curve observations. Source metadata and MD5 checksums are retained alongside SHA-256 hashes. Efficiency, uncertainty and delay axes are mapped explicitly. Separate heralded-photon arrays remain separate observables: a source time of 100 μs versus an article value of 100.9 μs is an unresolved clock-offset issue, not corrected by guessing.

The reference branch contains ten primary-text-reviewed values. It covers five Eu/YSO effective coherences, three vanadium-complex phase-memory values and two integrated-SiC coherence values. It is a seed reference table, not an exhaustive or current-best material atlas. Separately, a pinned molecular dataset [8] contains 201 positive entries across 123 source rows and 40 original-paper links. Missing uniform units, field and pulse conventions prevent their promotion to validated SI lifetimes. The primary-text review of ten values does not validate all corresponding secondary-dataset rows at other temperatures.

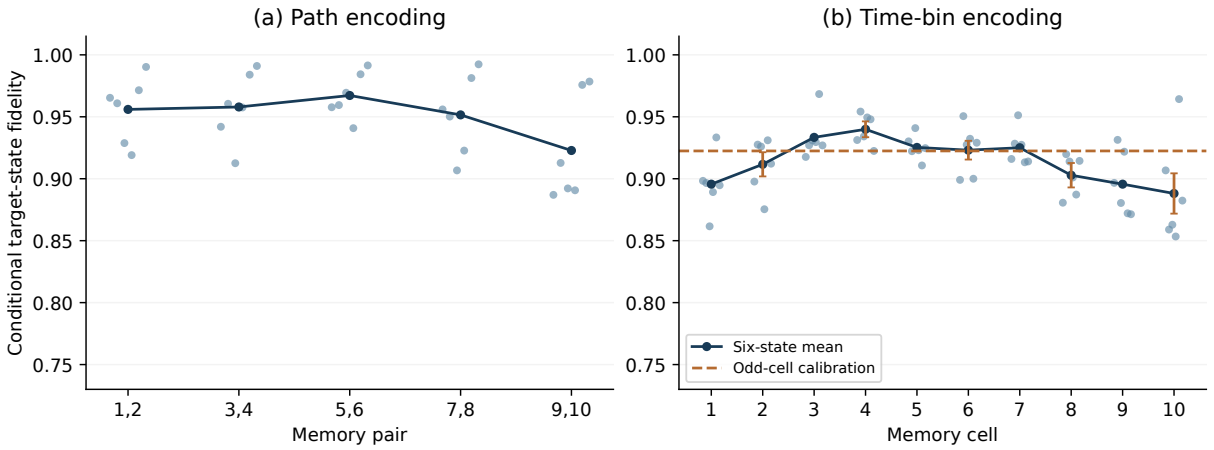


Figure 2: Reanalysis of processed data from Teller et al. [2]. Small points show the six state fidelities; dark markers show their uniform means. In panel (b), the dashed line is the inverse-variance calibration on odd-numbered cells, and error bars on even cells show supplied MC uncertainties. Conditioning, encodings and the spatial holdout are distinct. Error bars do not include shared systematic uncertainty.

Three deterministic specification cases exercise Eq. (4) and the finite bath. The count control uses 21 ages, four analysis phases per age, 4000 trials per phase, known $T_2 = 1$ ms and seed 20261010. Its 336,000 synthetic Bernoulli trials are not measurements. Sixty bootstrap replicates are requested for this control and thirty per experimental curve model. Eight primary curve-model fits are counted separately from bootstrap and holdout refits.

Two independent numerical descriptions are compared: the analytical Markov/channel construction and QuTiP evolution or conversion. The Teller operating-point network workload uses four seeds, 100,000 attempts each, an 18 μs age and a 1 ms attempt interval. A separate hypothetical three-age control uses 2000 attempts. Both declare ideal loaded Bell pairs and ideal heralded loss. The original 75 automated tests were rerun with that archived source; later test environments and extensions retain their own counts and source identities. The companion Supplement details the scope, manifests and commands.

6 Results

6.1 Reanalysis of the memory-array data

The uniform path and time-bin means are 0.9510824229 and 0.9140479475, reproducing the paper's rounded summaries of 95% and 91%. Released per-state data reproduce the corresponding figure means with maximum discrepancies 2.22×10^{-16} and zero, respectively. A released output density matrix gives target-X fidelity 0.9576996483, agreeing with its reported-state counterpart to 2.70×10^{-12}. These results check released processed-data consistency, not raw coincidence reconstruction.

The odd-cell calibration gives $\widehat{F} = 0.9224361$ with an approximate calibration uncertainty 0.0038484. The maximum held-out standardized discrepancy is 2.3417, below the diagnostic threshold. Figure 2 shows that passing this check is compatible with visible cell variation. It does not establish microscopic material uniformity.

We also fit fidelity versus *detection-window width* at fixed storage age. A finite Gaussian pulse

with flat unpolarized background has

$$F(w) = \tfrac{1}{2} + (F_s - \tfrac{1}{2})\frac{\mathrm{erf}(w/(\sqrt{8}\sigma_p))}{\mathrm{erf}(w/(\sqrt{8}\sigma_p)) + bw}, \tag{20}$$

with consistent time units. Calibration uses 80 widths at or below 600 ns; assessment uses 91 widths at or above 800 ns; 40 intermediate widths are unused. The three maximum held-out pointwise discrepancies are 1.729, 2.638 and 0.723. The corresponding constant-fidelity diagnostics have discrepancies 5.547, 5.279 and 4.119 and are flagged as mismatches. One fitted signal fidelity reaches its upper bound. Nested windows share counts, so these are pointwise adequacy diagnostics rather than independent statistical validation; Eq. (20) does not estimate T_2.

6.2 A finite operating-point channel and its network control

For path pair 5,6 and a 270 ns detection window, the reviewed point has conditional six-state mean $F_0 = 0.9671968089$ and supplied MC uncertainty 0.0051670. The reported cryostat temperature is 3 K; the optical input-to-output time is 18 μs, comprising 10 μs AFC rephasing and 8 μs spin storage. These are distinct from the detection-window width.

The chosen isotropic approximation is

$$\Phi_p(\rho) = (1-p)\rho + pI/2, \qquad p = 2(1-F_0). \tag{21}$$

This is one effective model calibrated to the mean, not a channel inferred uniquely from process tomography. The derived signal-acceptance model is

$$\eta_0 = 0.16 \times 0.42 \times 0.50 \times 0.42 = 0.014112, \tag{22}$$

combining AFC efficiency, two-way optical transfer, demultiplexing and temporal acceptance. The propagated component uncertainty is 0.0015824 under independent component errors and exact temporal acceptance. Transfer measurements use classical light; acceptance uncertainty and covariance are unavailable. Fiber coupling, filters, detectors and analysis success are outside this loss model. The simulation therefore does not reproduce a measured complete single-photon network efficiency.

Analytical and QuTiP Choi matrices at this point agree to 1.21×10^{-17}. For two identical conditional depolarizing channels on an ideal Bell input,

$$F_B = \frac{1 + 3(1-p)^2}{4} = 0.9048185746, \qquad P_{\mathrm{pair}} = \eta_0^2 = 1.99148544 \times 10^{-4}. \tag{23}$$

SeQUeNCe produces 20, 32, 24 and 11 retrieved pairs for seeds 20261010–20261013, respectively: 87 in 400,000 attempts. The pooled fraction is 2.175×10^{-4} with Wilson 95% interval $[1.76350, 2.68249] \times 10^{-4}$, containing the fixed-loss expectation. Maximum joint-density discrepancy from the analytical result is 1.11×10^{-16}. Figure 3 reports Monte Carlo uncertainty only. The same assumptions give a simulated rate 0.2175 s^{-1} over 400 s of combined attempt time; this is not a measured network rate or a validation of its missing optics.

An additional constructive calculation makes the mean-only limitation explicit. A phase-flip channel with probability $q = 3p/4$ has the same average fidelity as Eq. (21), but two such channels yield a different Bell fidelity. The Supplement records the exact matrices and numerical values. This example concerns the mean constraint alone; it does not assert that both models fit every released state or density matrix.

6.3 Failure of simple decay models on the separate Pr experiment

Figure 4 shows the released efficiency curves and the two tested envelopes. All six RF-curve/model combinations are flagged as model mismatches under the declared Gaussian-error diagnostic. XX

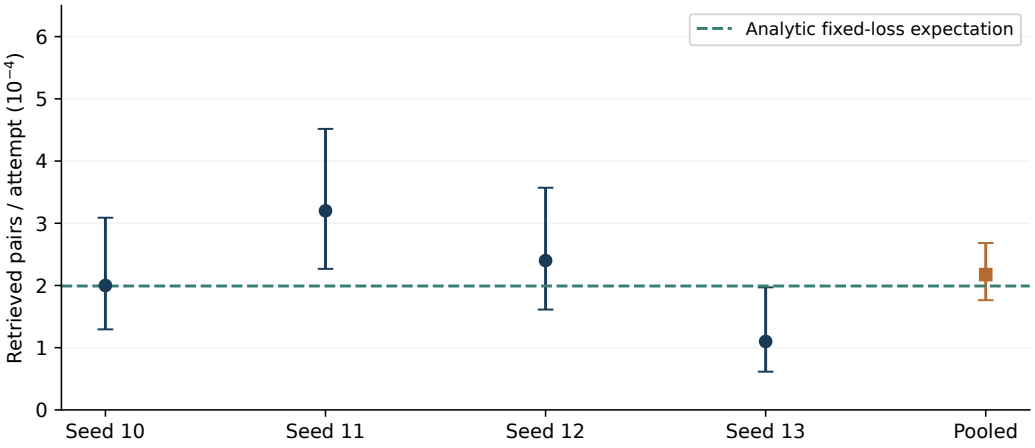


Figure 3: Ideal-input network integration control for the calibrated, nonunique Teller operating-point model. Individual seeds each use 100,000 attempts; the pooled point uses 400,000. Bars are Wilson intervals from retrieval counts under fixed parameters. Agreement with the analytical loss model verifies implementation; uncertainty in the optical parameters and false-herald physics is not included.

held-out RMSE values are 3.582 (exponential) and 2.853 (stretched); XY4 values are 3.786 and 3.434; XY16 values are 1.983 and 1.417. The latter two pass the held-out RMSE screen but fail the full-data goodness-of-fit diagnostic. The source paper itself discusses nonexponential behavior [4]; our analysis verifies that the simple diagnostic path does not turn those curves into an unsupported homogeneous lifetime.

The no-RF exponential fit returns an apparent scale outside its observed interval; the stretched fit has insufficient stable bootstrap draws. Both are inconclusive. No fit in this case returns an identified physical spin T_2. Reported tiny Gaussian goodness-of-fit probabilities condition on independence and the supplied error model; they are not evidence for a unique microscopic cause of the mismatch. Modulations, noise, field uncertainty and protocol details require additional modeling. The withheld heralded-photon arrays also do not measure qubit fidelity, so they cannot repair that missing identification by themselves.

6.4 A distinct-family nuclear-spin observable case

The additional experimental case uses Bradley et al.'s Fig. 3b and inset for a ^{13}C spin associated with an NV centre in isotopically engineered diamond [14, 15]. Storage is free Ramsey evolution in heralded NV^0 at 4 K, followed by recharging and NV$^-$ postselection; no refocusing pulses are used. The reviewed field is $(0.3 \pm 0.1, 0.06 \pm 0.08, 46.801 \pm 0.001)$ G in the NV frame. The data include 13 transverse Bloch-length points and 11 signed X points with reported errors, without SPAM correction. Times are converted from milliseconds to seconds; no artificial zero-time point, unit normalization or clipping is introduced.

The physical context is $G = \nu_L I_z + m_s(A_\parallel I_z + A_\perp I_x)$. Hyperfine values measured in NV$^-$ do not independently identify the NV^0 tensor or its noise process, so they are not used as a parameter-free prediction. Instead, the executed effective model uses $G_{\text{eff}} = \Delta\nu I_z$, $I_z = \text{diag}(1/2, -1/2)$, with $\Delta\nu = 79.8$ Hz fixed from a separate NMR measurement and the RF reference, not fitted to held-out Ramsey points. For $E(t) = \exp[-(t/\tau)^\beta]$,

$$\ell_{\perp,\text{model}}(t) = AE(t), \qquad X_{\text{model}}(t) = E(t)[c\cos(2\pi\Delta\nu t) + s\sin(2\pi\Delta\nu t)]. \tag{24}$$

Measurement amplitudes A, c, s remain separate from the normalized channel dephasing. Independent matrix-exponential propagation verifies the specified rotation/channel equations.

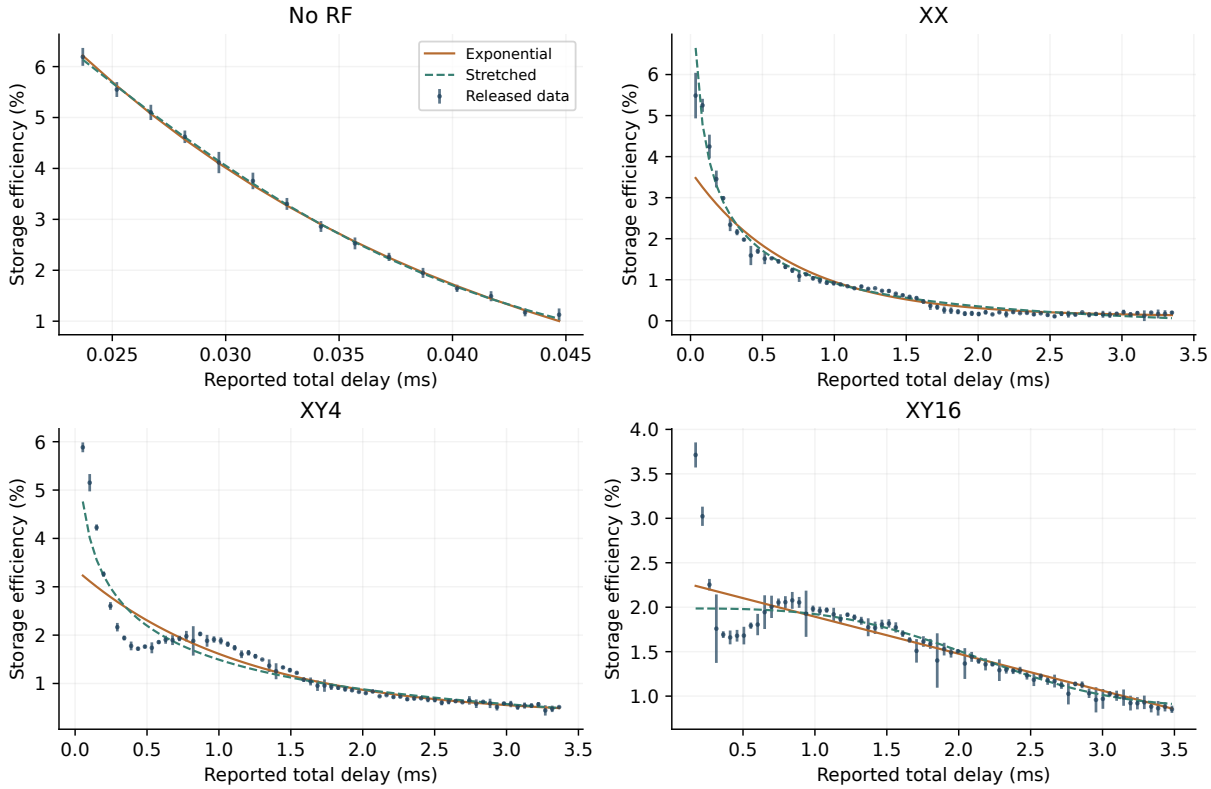


Figure 4: Processed open classical-efficiency data [5] and diagnostic fits with Eq. (17). Three RF curves have 70 observations each; the no-RF curve has 15. Curves are plotted against the source's reported total delay. Error bars are the released uncertainties. Fits are apparent retrieval envelopes, not inferred spin-T_2 curves. The panels have different horizontal scales.

The first nine envelope points calibrate A, τ, β and the last four are held out; the first five X points calibrate only c, s and the last six are held out. This split was chosen after source inspection and is not prospective independent validation. The fit gives $A = 0.53533$, $\tau = 55.247$ ms, $\beta = 1.36263$, $c = 0.48286$ and $s = 0.02318$. This τ is an effective subset-fit scale without a new physical-T_2 confidence interval.

The maximum held-out absolute residual divided by its source error is 2.428 for the envelope and 3.158 for X, against the fixed threshold three. The X failure persists at 79.7 and 79.9 Hz. Calibration uncertainty, interpoint covariance and raw-shot likelihood are unavailable, so this is a descriptive consistency screen rather than a calibrated significance test. The observable comparison is completed with its negative result retained, while microscopic physical validation and optical network export remain false. This broadens the experimental data examined to a defect-spin memory; it does not complete the original strict, independently parameterized pure-dephasing baseline.

6.5 Deterministic controls and condition-sensitive references

For the supplied Markov control, $\Gamma_1 = 10\ \mathrm{s}^{-1}$ and $\Gamma_\phi = 20\ \mathrm{s}^{-1}$ give $T_2 = 40$ ms at its declared hypothetical conditions. Independent parameter-box ranges $[8, 12]$ and $[18, 22]\ \mathrm{s}^{-1}$ produce $T_2 \in [35.714, 45.455]$ ms. Four corner evaluations implement this range. Its relative deviation is $-10.71\%/+13.64\%$ about the nominal value; this is deterministic interval propagation, not prediction accuracy or a statistical confidence interval.

The one-nucleus bath control specifies $I = 1/2$, central projections $\pm 1/2$, longitudinal hyperfine coupling $A_{zz} = 4$ kHz and $T = 4.2$ K. For the high-temperature unpolarized limit, $L(t) =$

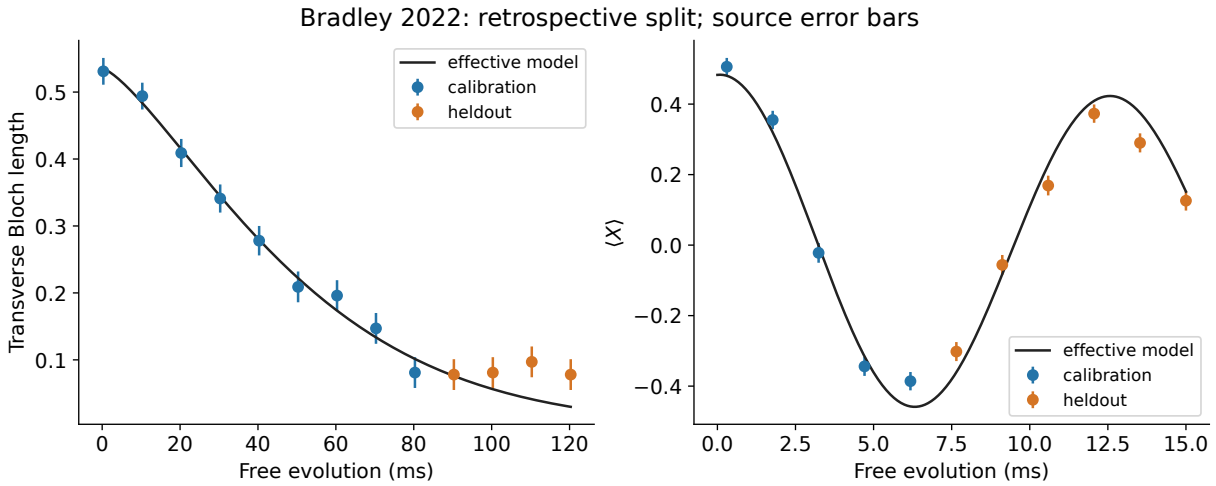


Figure 5: Bradley Fig. 3b reanalysis [15]. Calibration and held-out points are separated; the two measured observables use different time axes and measurement amplitudes. Source errors are unchanged. The envelope passes the descriptive residual screen, while signed X fails. The positive Bloch length can contain a noise floor not modeled here; this result does not overturn the original experiment's interpretation.

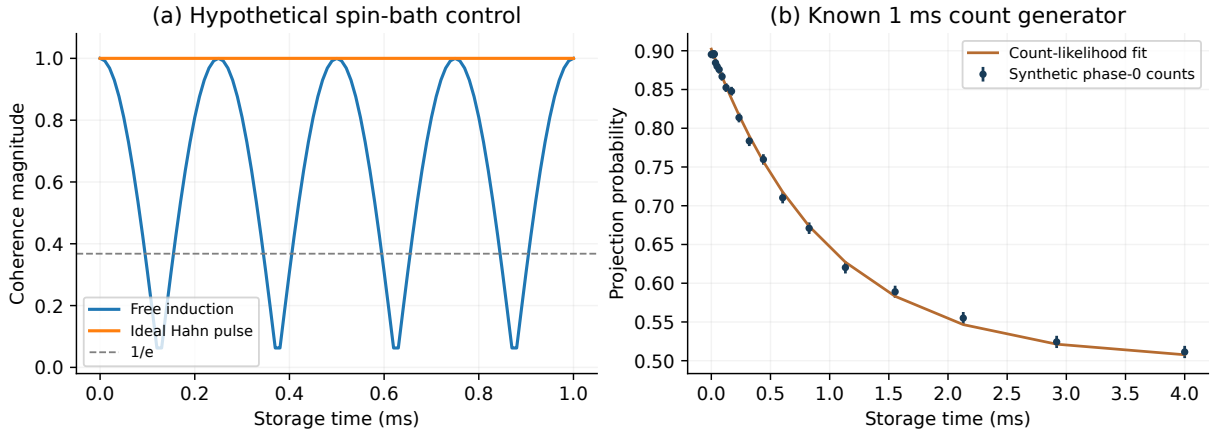


Figure 6: Explicitly synthetic verification cases. (a) Supplied one-nucleus Hamiltonian under free evolution and ideal Hahn refocusing; a finite bath can revive and need not have an exponential lifetime. (b) Phase-zero projections from the four-phase binomial control and fitted probabilities. All phases, rather than only the displayed phase, enter the likelihood.

$\cos(\pi A_{zz} t)$. The finite-temperature implementation uses the Gibbs state exactly. Free evolution has revivals and a first sampled $1/e$ crossing bracket of 90–100 μs. An ideal Hahn pulse refocuses the commuting longitudinal coupling, with no sampled crossing up to 1 ms. These are software controls, not predictions for a newly selected molecule.

The count control recovers 0.979663 ms from the known 1 ms generator, with a conditional parametric-bootstrap 95% interval of [0.950215,1.007737] ms. Recovery in one synthetic instance checks the likelihood implementation; it is not a coverage study or evidence for physical prediction accuracy. All phases at a held-out age remain outside the holdout fit.

The reviewed table query returns the 230 ms Eu/YSO XY16 effective-spin reference under its reported matching conditions [3]. An unknown-material query yields no applicable reference; changing a required temperature also prevents a match. Ten reviewed rows and 201 provisional molecular observations coexist without sharing the same physical-usability status. Three deter-

ministic specifications, 202 bath-time evaluations and four rate-box corners are recorded, alongside the eight experimental curve-model fits.

The separate hypothetical network control runs SeQUeNCe for 2000 attempts and retrieves 321 pairs. QuTiP/analytical full-channel Choi difference is 4.00×10^{-15} and maximum retrieved-density discrepancy is 5.55×10^{-17}. These are checks of supplied write/store/read parameters, distinct from the experiment-constrained point above. The original automated suite passes 75 tests, covering units, channels, bath dynamics, adapters, source handling, reference restrictions and workflow behavior.

7 System evaluation: executed steps 1–4

These additional evaluations address a matched computational reference, a second-family case pathway, controlled contract ablations and local execution performance. All four pathways were exercised, but the physical objective of step 2 was not completed: its executed molecular-spin case is synthetic. Table 2 and the Supplement distinguish mathematical/software verification from experimental evidence. Configuration and source hashes, actual seeds and failure statuses are retained in a separate system-evaluation claim map. The four selected records share one evaluation-source fingerprint; confirmations of steps 2 and 3 use the published evaluation implementation. No machine-learning training or new hardware experiment is included.

7.1 Step 1: matched direct-script comparison

A separately written reference imports NumPy/SciPy without importing the lab's channel-construction routines. It builds the column-vectorized Lindblad generator and evaluates its matrix exponential, then independently forms the conditional qubit map and the erasure-inclusive qubit-to-qutrit Choi matrix. Both arms receive identical input conditions and conventions. Six Pauli eigenstates, full Choi matrices, conditional average fidelity and state-independent acceptance are compared with tolerance 10^{-8}. Both source parsers check the same raw SHA-256 hashes; the direct arm is not deprived of source-integrity checks.

The workload has the calibrated Teller operating point and twelve hypothetical Markov profiles: four joint choices of population rate, detuning and equilibrium population, each at three storage temperatures and six ages. These are 73 channel-age points, not 73 independent experiments. Two independent parsers also reproduce the same 90 released processed fidelities, with zero discrepancy. Maximum full-Choi, six-state output and average-fidelity errors are 1.3103×10^{-14}, 1.1828×10^{-14} and 3.9968×10^{-15}, respectively; acceptance agrees exactly.

Quantity	Result
Profiles / channel-age points	13 / 73
Processed fidelity values	90
Maximum full Choi discrepancy	1.310×10^{-14}
Maximum six-state output discrepancy	1.183×10^{-14}
Maximum conditional average-fidelity discrepancy	3.997×10^{-15}
Maximum acceptance / processed-value discrepancy	0 / 0
Declared numerical gate	1.000×10^{-8}

Table 2: Step 1: matched numerical and processed-data comparison. Twelve profiles are supplied software controls; one profile is an experiment-constrained point.

Five randomized-order warm construction repeats per arm, with seed 20261010, give medians 0.054380 s for the lab and 0.099634 s for the direct implementation. These timings do not equalize metadata, export, validation or metric-construction work and exclude source retrieval,

file IO, fitting and network events. They are descriptive local measurements, not evidence of end-to-end speed superiority or saved researcher effort. AiiDA already provides provenance and caching [18]; it was assessed through documentation, not run as a comparator. Equivalent domain contracts could be implemented in custom scripts or workflow-engine plugins. The matched result verifies numerical preservation across the supplied integration boundary, rather than establishing superiority over existing frameworks.

7.2 Step 2: second-family pathway and its unresolved physical baseline

The case runner accepts a hashed complex pure-dephasing curve, a separately supplied spin Hamiltonian and reviewed temperature, field/frame, isotope/composition and pulse conditions. It compares a fixed model with the curve without fitting parameters. A total echo envelope cannot be silently relabeled as pure dephasing; optical write/read review is a separate requirement. Missing optical information permits a coherence comparison but blocks network export. An unsuccessful curve gate also blocks export. Hypothetical inputs require explicit network opt-in and retain hypothetical evidence.

The executed control uses one supplied ^{13}C spin at 4.2 K, longitudinal hyperfine $A_{zz} = 4000$ Hz, nuclear Zeeman frequency $\nu_z = 214163.9829$ Hz and no pulses. The 41 ages span 0–1 ms. Its independent closed-form curve is

$$L_{\mathrm{ref}}(t) = \cos(\pi A_{zz} t) - iP \sin(\pi A_{zz} t), \qquad P = -\tanh\left(\frac{h\nu_z}{2k_B T}\right). \tag{25}$$

The declared complex-residual scale is 0.01 and the gate is $\max_t |L_{\mathrm{model}} - L_{\mathrm{ref}}|/0.01 \leq 3$. This scale is chosen for a software control, not a measured SEM. The maximum scaled residual is 3.72×10^{-12}. Revivals occur; a first sampled $1/e$ crossing is bracketed by 75–100 μs on this coarser grid. Neither this bracket nor the residual is a statistical T_2 prediction error.

With explicitly hypothetical optical parameters, the provider-neutral two-node workload uses 10,000 attempts, herald probability 0.1, age 500 μs and seed 20261010. It gives 1020 heralds, 313 retrieved pairs and conditional Bell fidelity 0.9559176317. This is the lab's channel benchmark, not another SeQUeNCe or NetSquid run. A molecular reference [6] is retained as a prospective physical case, but its missing reviewed curve and Hamiltonian produce a blocked status. Consequently, this original step-2 control establishes software-path consistency only. The subsequent distinct-family observable comparison in Section 6.4 uses measured data with calibrated effective dynamics, retains a failed X diagnostic and does not complete the strict physical-model baseline.

7.3 Step 3: constructed contract ablations

Seven selected invalid inputs test age and temperature extrapolation of a measured point, reuse of a total coherence rate as pure dephasing, unreviewed noise independence, hypothetical network use without opt-in, incremental use of an absolute-age map, and modified source bytes. Six protections belong to the existing lab; the source-hash contract in this suite belongs to the evaluation layer. Each has a passing positive control and an isolated deliberately unsafe bypass; no production guard is modified. All seven invalid cases are rejected and all bypasses accept the constructed cases.

The consequences have distinct units. Reusing a total coherence rate of 25 s^{-1} as pure dephasing while also adding $\Gamma_1/2 = 5$ s^{-1} changes the specified 40 ms to 33.333 ms, a 16.667% conditional error. Applying a complete age map twice gives maximum density-element error 0.097247 in the chosen control. Omitting the ms-to-s conversion produces maximum density-element discrepancy 0.132201 in an additional unit control. Other rows count unsupported conditions or evidence promotion, rather than physical prediction errors. The Supplement reports each case separately. This selected fault coverage does not estimate researcher error incidence or establish that another framework cannot implement equivalent guards.

7.4 Step 4: local performance, checkpoint reuse and robustness

The native study runner evaluates two supplied models (Markov and finite bath), three detunings and two replicates on six ages, giving 12 jobs per cold study. Worker counts 1, 2 and 4 are ordered randomly for each of three repeats using seed 20261010. Each cold study has a separate output directory; a subsequent warm pass reuses its terminal-result checkpoints. Native numerical threads are limited to one per worker. Nine cold studies complete 108 jobs; nine warm passes reuse 108 checkpoints. Configuration identities and serialized channel results match across all worker counts and repeats.

Workers	Cold median [min,max] (s)	Resume median (s)	Median jobs/s
1	3.8674 [3.8089, 3.9101]	0.01814	3.103
2	1.9781 [1.9165, 2.0665]	0.01741	6.066
4	1.1782 [1.1299, 1.2286]	0.01817	10.185

Table 3: Step 4: three local cold studies and three checkpoint-reuse passes per worker count, with 12 jobs per study. Bounds are observed minima/maxima, not confidence intervals.

Cold medians decrease from 3.8674 s at one worker to 1.9781 s at two and 1.1782 s at four. These bounded local results include process startup, calculation and artifact IO, and do not establish cluster-scale or cross-machine scalability. Parent peak RSS is 34.578–37.078 MiB, and the largest terminated child's peak is 63.238–66.414 MiB; these maxima are separate measurements, not aggregate concurrent memory. Missing-parameter and real process-timeout probes yield failure statuses `missing_data` and `timeout`. Restart after one normally completed checkpoint completes the study with exactly one resumed result. No forced-crash, power-loss or filesystem-corruption test was performed.

A fresh GitHub Actions checkout at the recorded release commit installs the core package, passes seven evaluation tests and retrieves the two pinned experimental files automatically before rerunning step 1. Its maximum full-Choi discrepancy is 1.3214×10^{-14}, also below the same gate. This checks the core clone/install/retrieval path; it does not replace full optional-backend reproduction or colleague-led physical and scientific review.

8 Joint protocol, convergence and paired network validation

The extension follows the order instrument verification, bounded spin-bath convergence, paired simulator pilot and larger simulator study. Numerical prerequisites are checked before the next workload starts. These gates authorize the following *software control*, not an experimental or new-material prediction. All configurations and thresholds are retrospective; no new hardware measurement is included.

8.1 Time-dependent fidelity, signal retrieval and readout noise

Seven hypothetical instruments cover identity, complete loss, background-only acceptance, dephasing with loss, thermal/background noise, biased background and noise-dominated retrieval. Eight storage ages span 0–100 ms. A separately implemented Liouvillian and matrix-unit reference constructs the complete qubit-to-qutrit map in Eq. (14); it does not reuse the lab's Kraus construction. The checks include the full Choi matrix, six Pauli-state outputs, average fidelity, signal/noise/acceptance probabilities, complete positivity and trace preservation. All 56 channel-age points and 336 state probes pass at tolerance 10^{-10}; the maximum numerical discrepancy is 1.443×10^{-15}.

An audit of the bundled PR2026 heralded-photon arrays retains nine ages with efficiency,

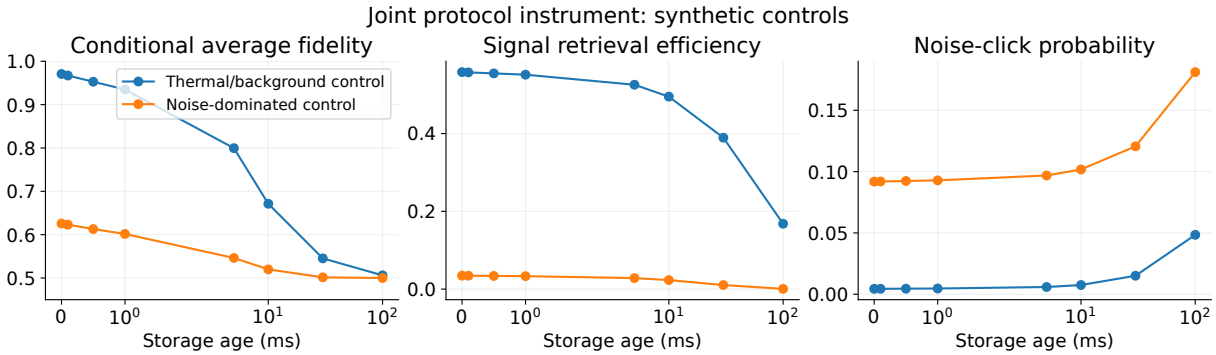


Figure 7: Joint instrument controls, generated from retained numerical results. Signal efficiency, accepted noise and conditional fidelity vary together under the supplied rates. Background-only clicks do not become signal efficiency; a noise-dominated accepted output can approach average fidelity 1/2. Curves are synthetic effective models, not experimental fits. The horizontal scale is linear near zero and logarithmic at longer ages.

Control	Final comparison	Maximum error	Tolerance
Matrix exponential / spectral	4-spin reference	1.444×10^{-15}	1.000×10^{-10}
Strang substeps	128 substeps	2.332×10^{-5}	1.000×10^{-4}
CCE order	Order 4; 4 spins	4.132×10^{-10}	1.000×10^{-3}
Pair cutoff	40 angstrom; 4 spins	4.132×10^{-10}	1.000×10^{-3}
Bath size	6 to 8 spins	1.452×10^{-8}	1.000×10^{-3}
Mean $\lvert L \rvert$ integration	129 to 257 points	1.950×10^{-6}	1.000×10^{-4}

Table 4: Final convergence diagnostics over 0–0.4 ms, taking the maximum over 0, 1 and 4 ideal pulses. Solver, CCE and cutoff errors use the exact four-spin reference; bath-size and time-grid errors compare adjacent nested levels. The time-grid metric is mean absolute coherence, not every revival extremum or an inferred lifetime.

noise and signal–idler $g^{(2)}$, together with source errors. They have no matching qubit-fidelity series. The 0.9 μs source/article clock difference remains unresolved; noise per herald cannot be assumed to equal a vacuum false-click probability. Neither Teller's fixed operating point nor Bradley's nuclear-spin observables supplies the missing optical time series. Joint *experimental* validation therefore remains inconclusive, with no fidelity inferred from $g^{(2)}$ or imported from another device. The pending data contract requires matched state/tomography, retrieval and background-only counts, common trial denominators and clocks, detector windows, photon statistics, pulse/field/temperature conditions, and uncertainty/covariance definitions.

8.2 Bounded noncommuting spin-bath convergence

The nested synthetic bath has 2, 4, 6 and 8 ^{13}C spins, maximally mixed initialization and central projections $m = \pm 1/2$. Noncommuting transverse hyperfine terms accompany longitudinal couplings and explicit nuclear-pair tensors. Ideal control has 0, 1 or 4 CPMG pulses over 0–0.4 ms. An independent spectral propagator is compared with matrix exponentials, followed by Strang-step, CCE-order, pair-cutoff, bath-size and time-grid studies. There are 69 actual coherence solver calls and 66 diagnostic rows; final comparisons pass all six controls (Table 4). Lower-level failures remain in the archived table and Fig. 8.

The supplied outer couplings deliberately decrease rapidly. This is not a material lattice, measured hyperfine distribution or disorder ensemble. Consequently, the small last-step error cannot authorize a larger bath or longer observation window for another material. PyCCE now receives all explicitly supplied pair tensors, including zero tensors where relevant; otherwise it

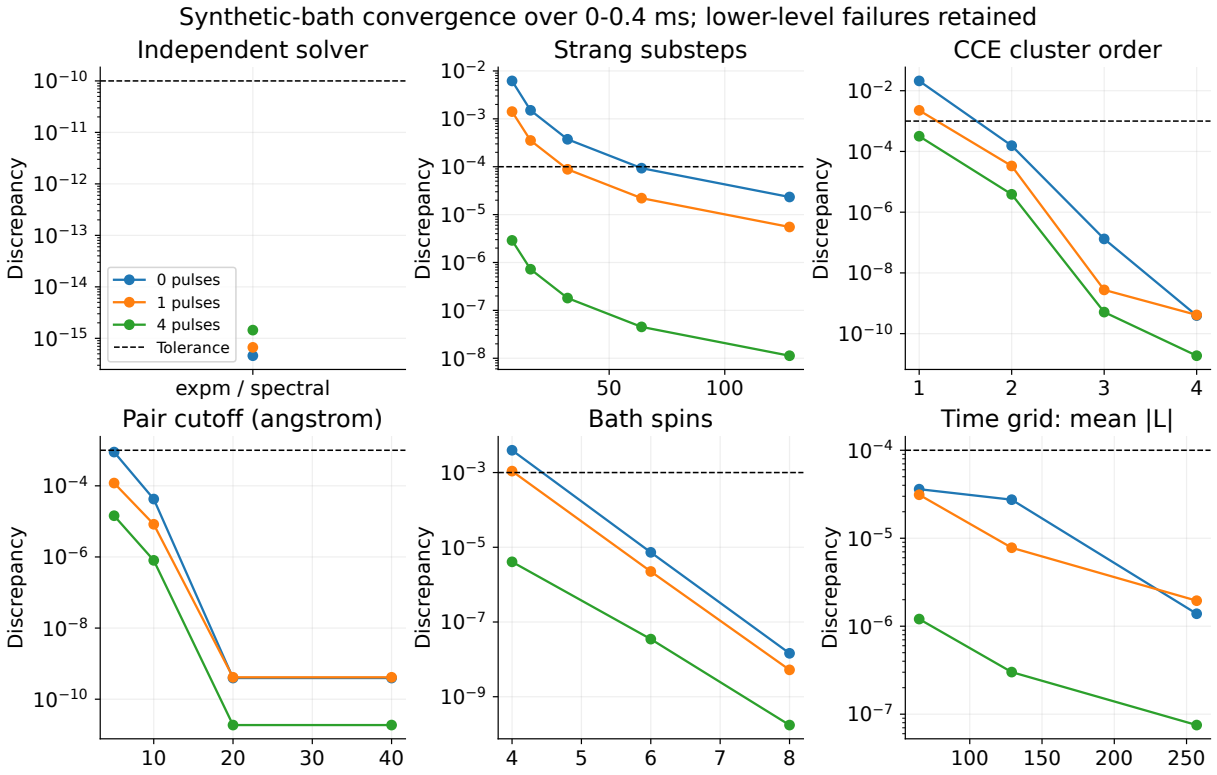


Figure 8: All six convergence controls for the supplied noncommuting synthetic baths; dashed lines are declared numerical tolerances. Bath-size and time-grid points compare adjacent levels and omit the first level, which has no preceding comparator. Other panels use the independent/exact four-spin reference. The time-grid ordinate is an error in integrated mean $|L|$. Ideal-pulse propagation convergence is not a pulse-shape or pulse-error study.

can generate dipolar interactions for unspecified pairs. The field/isotope Zeeman convention is checked against the exact reference. The Supplement records the full Hamiltonian recipes and unit conversions.

Profiles explicitly declaring `prediction_purpose: material_prediction` now require a passing certificate with a matching profile-content hash, five converged refinement axes, error below the declared tolerance, a valid time interval and independent physical-model review. Hypothetical evidence cannot authorize such a prediction. This checks declared evidence and identity, not the scientific validity of a self-issued certificate; legacy profiles without that purpose label continue to run without receiving a new material endorsement. No material certificate is issued by this study.

8.3 A second native simulator and the larger paired study

SeQUeNCe 1.2.0 runs its native event kernel and density manager. QuNetSim 0.1.2 [16] runs its native hosts, quantum connections, packet queues and acknowledgements with a new protocol-specific QuTiP density backend. The latter preserves joint states during transmission and consumes retrieved/lost qubits; it is not QuNetSim's stock EQSN backend. An independent matrix-unit reference checks local and joint channel action. These are two executed simulator stacks, rather than two channel descriptions in one simulator.

The stacks share external logical ages and three separate seeded streams for heralding and the two readouts. Sixteen parameter combinations use background rates 0 and 5000 s^{-1}, ages 0, 1, 10 and 30 ms, and herald probabilities 0.25 and 0.75. Attempts are spaced by 40 ms and the model ensemble Bell-fidelity gate is 0.8. Quality qualification uses this ensemble gate, not

Study	Seeds	Pairs of runs	Arm runs	Trials/run	Trials, both arms
Pilot	3	48	96	32	3 072
Larger	8	128	256	512	131 072

Table 5: Actual native simulator work counts. Each parameter/seed combination is executed once per simulator. Paired shared traces test implementation agreement; seeds are the independent Monte Carlo clusters. Pilot and larger stages are separate processes and use different seeds.

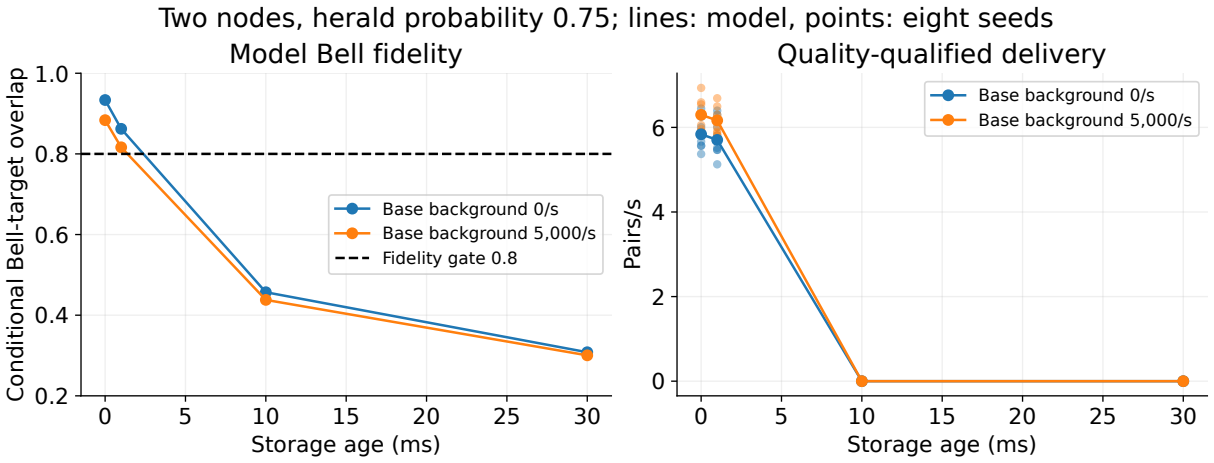


Figure 9: Larger-study ensemble Bell fidelity and quality-qualified pair rate at herald probability 0.75. Lines show the supplied model expectation at sampled ages; points show all eight seed-cluster outcomes. Acceptance can increase with background while fidelity decreases, so accepted-pair counts alone are insufficient. Eight seeds are eight independent Monte Carlo clusters per parameter combination; no cross-parameter independence, coverage, power or speedup is inferred. Both simulators give the same paired counts.

unobservable per-trial signal/noise labels. The pilot and larger study use separate fresh processes because QuNetSim's global network instance is not restarted within a completed worker.

After the pilot passes, the larger study completes 256 arm runs and 131 072 trials across both simulators. All 128 paired comparisons have identical branch counts and logical trace hashes; the maximum difference between mean joint density matrices is 2.465×10^{-15}, below 10^{-10}. SeQUeNCe executes 98 344 native events and QuNetSim transfers 32 808 quantum packets. No arm fails. Paired runs have identical random inputs by design; their agreement is implementation verification, not independent experimental replication.

QuNetSim uses real-time worker scheduling [16], which is not equated with a physical network clock here. The common logical trace is an externally imposed test input; agreement of its hash does not cross-validate native physical latency. The workload has two nodes and one slot, and omits routing, swapping, contention, finite fiber propagation and detector double clicks. This boundary is narrower than latency-and-resource cross-validation such as Chung et al. [19]. Runtime records are retained as execution metadata without a comparative performance claim.

8.4 Replay and current test scope

A fresh denied-network replay reproduces 359 JSON records and 11 exact random-trace arrays, including every archived native run. Lab/evaluation/configuration fingerprints match; the absolute and relative numerical tolerance is 10^{-10}. Each ZIP archive's SHA-256 is verified before its contents are compared. Timestamps, environment/runtime metadata and archive-container bytes are excluded from scientific equality. The current extension environment runs 109 tests: 107 pass

and 2 optional PySCF/MaxwellBloch tests are skipped. These counts describe that environment; they do not replace the earlier full-environment test snapshot. Historical system stages 1–4 also preserve their full scientific payloads under an explicit cross-revision comparison. All checks are author-operated, not colleague-led independent scientific review.

9 Discussion

The executed results support the framework's ability to preserve distinctions that matter physically. It reproduces published processed-data summaries, constructs a conditional point model with explicit ambiguity, carries that model into an actual event simulator, and withholds physical lifetimes when its data or model are insufficient. The negative outcomes are part of the evidence: the constant readout-window model is inadequate, the simple RF-decay models mismatch, and an unknown material has no table-derived lifetime.

The results do not validate an arbitrary chain from electronic structure to phonon relaxation to optical memory fidelity. Such a chain needs experimentally anchored Hamiltonians, bath and phonon parameters, pulse errors, sample thermometry, optical coupling and a detection/noise instrument. Reproducing a processed mean does not reconstruct raw trials; a spatial holdout within one dataset does not supply independent hardware replication; numerical Choi agreement does not compensate for omitted physics. Several reference values come from each study, and many molecular entries share source papers, so row counts should not be interpreted as independent sample sizes.

The main implementation boundary is useful even for negative candidate outcomes. Structural or charge/isotope variations can be enumerated and supplied to the appropriate solver. If an optical interface is unknown, coherence can be studied under explicit assumptions while device usability remains unresolved. If the dynamical model shows revivals or a short-window fit is unidentifiable, the output can preserve the curve or bounds instead of forcing a single rankable number. No chemical combination is labeled experimentally new merely because it is absent from the local database.

There are also limitations at the software level. The family taxonomy is finite, bath dimension is bounded, classical propagation is not automatically promoted to a single-photon instrument, and exported histories have fixed control and exact-age semantics. State-dependent heralding, intervention-dependent non-Markovian histories, thermal ramps and realistic generation/detection require new validated contracts. SeQUeNCe and QuNetSim now supply a bounded two-stack state/trace comparison, but their physical network timing is not cross-validated. NetSquid and QuISP were not run. The additional local benchmark exercises small models and worker counts only; runtime and scaling on a cluster or supercomputer were not benchmarked. The strict second-family physical-model baseline remains incomplete: a subsequent measured Bradley case adds an observable comparison with a retained X failure, rather than an independent microscopic validation. Adapter availability should therefore not be mistaken for validated microscopic coverage of all elements and molecules.

Next validation should use time-resolved fidelity, success and noise at the same storage times, with raw counts, preparation/analysis settings, photon statistics and covariance. Realistic spin-bath applications need independent parameter review and convergence over bath size, cluster order, pair cutoff, control resolution and time grid for each material and domain. The supplied synthetic convergence results cannot establish that transfer. Further paired network benchmarks must test native physical timing, multiple slots, swapping and queue contention separately from the imposed logical trace used here. These additions would broaden the system's validated scope, regardless of whether a new candidate improves performance.

9.1 Historical clean-clone repair and numerical replay

A subsequent v0.6.1 repair addressed reproducibility failures in a fresh public clone. Seventeen unchanged, licensed measurement files are bundled with attribution and content hashes. The default paper replay recomputes the numerical analyses and network controls offline into a separate result tree, while retaining reviewed paper transcriptions as inputs. It excludes reacquisition of the molecular collection and reverification of unbundled primary-paper bytes. The historical 201-entry source audit is therefore not a fresh offline result.

An author-operated clean clone with isolated Python 3.12 core/full environments was tested with external DNS and sockets denied after installation. The core suite passes 68 lab and 18 evaluation tests with seven optional skips; the full environment passes all 93 tests. Two system passes each execute 108 cold jobs and 108 checkpoint reuses. All four full scientific-payload comparisons pass at absolute/relative tolerance 10^{-10}. Ten reporting checks pass; five PDF/PNG figure pairs regenerate identically. The strict physical second-family descriptor remains blocked. These are reproduction controls, not a new performance comparison or independent scientific review. The Supplement and repair dossier retain methods, source identities and all twelve reviewed findings.

10 Reproducibility and availability

The public repository includes source, configurations, licensed measurement inputs, results, figure/table scripts, claim mappings and this Supplement. External datasets retain their own DOIs [2, 5, 15] and license notices. Unlicensed third-party caches remain excluded from Git. The original snapshot was reproduced from the software archive with SHA-256

`c64b94a573def9d999b65be7201947e9c`
`2ce4e496aaf201650d7c1e3d44733aa`

Bradley and the validation extension retain separate execution/source identities. Source and article materials are available at `https://github.com/VesterlundCoder/memory-discover-lab`. The historical core-CI record identifies commit `de52b32`; each subsequent result identifies its own source/configuration fingerprints. No software DOI or peer-review acceptance is asserted.

The archived full environment uses NumPy 2.5.3, SciPy 1.18.1 and PySCF 2.14.0; the extension uses NumPy 2.3.5, SciPy 1.16.3 and QuNetSim 0.1.2, without PySCF/MaxwellBloch. Both record Python 3.12.14, PyCCE 1.1.1, QuTiP 5.3.1 and SeQUeNCe 1.2.0. Exact runtime pins are retained per revision. The extension ran with uncommitted source changes on base `985c8fe`; its actual lab/evaluation content hashes identify the executed implementation, subsequently published at `9358173`. `scripts/document_article_update.py` checks its retained claims and generates number macros, two tables and `evidence/article_update_claim_map.json`. PDF compilation is separate from numerical replay. The Supplement supplies exact commands, seeds, compatibility details and evidence pointers. AI-assisted coding and drafting were used; scientific responsibility remains with the named author.

11 Conclusion

WestQuant Memory Discovery Lab links versioned source data to scoped memory observables and network instruments. Experimental reanalysis retains both agreement and model mismatch; the distinct-family Bradley case adds measured spin observables while preserving its Ramsey-X failure. Independent references, bounded synthetic-bath convergence and paired native-simulator runs verify specified numerical contracts. Matched experimental fidelity/efficiency/noise, independently identified material dynamics and physical network latency remain unresolved. The contribution is a reproducible integration workflow with explicit evidence boundaries.

References

[1] M. Teller, S. Plascencia, S. Grandi and H. de Riedmatten, "Quantum Storage of Qubits in an Array of Independently Controllable Solid-State Quantum Memories," *Physical Review X* **15**, 031053 (2025). doi:10.1103/z6lc-qw2d.

[2] M. Teller et al., "Supporting data of Quantum storage of qubits in an array of independently controllable solid-state quantum memories," Zenodo record 14929997 (2025). https://doi.org/10.5281/zenodo.14929997.

[3] A. Ortu, A. Holzäpfel, J. Etesse and M. Afzelius, "Storage of photonic time-bin qubits for up to 20 ms in a rare-earth doped crystal," *npj Quantum Information* **8**, 29 (2022). https://doi.org/10.1038/s41534-022-00541-3.

[4] E. Rodríguez-Moldes et al., "Long-Lived Telecom-Heralded Single-Photon Storage in an Absorptive Spin-Rephased Quantum Memory," *Physical Review Letters* **137**, 120803 (2026). https://doi.org/10.1103/ftkb-pkvp. Author manuscript: arXiv:2609.02579v1.

[5] E. Rodríguez-Moldes et al., "Dataset from Long-lived telecom-heralded single-photon storage in an absorptive spin-rephased quantum memory," Zenodo record 21532904 (2026). https://doi.org/10.5281/zenodo.21532904.

[6] C.-Y. Lin, T. Ngendahimana, G. R. Eaton, S. S. Eaton and J. M. Zadrozny, "Counterion influence on dynamic spin properties in a V(IV) complex," *Chemical Science* **10**, 548–555 (2019). https://doi.org/10.1039/C8SC04122A. Primary full text: PMCID PMC6335635.

[7] K. R. Mullin, D. Johnson, D. E. Freedman and J. M. Rondinelli, "Systems-chart approach to the design of spin relaxation times in molecular qubits," *Dalton Transactions* **53**, 16585–16591 (2024). https://doi.org/10.1039/D4DT02311K.

[8] MTD-group, *Molecular Qubit Dataset*, source revision 5263b1d86b39243e7405dde5bf7c58bbd8fcd677, accessed 10 October 2026. https://github.com/MTD-group/Molecular_Qubit_Dataset.

[9] H. Hu et al., "Room-temperature waveguide integrated quantum register in a semiconductor photonic platform," *Nature Communications* **15**, 10256 (2024). https://doi.org/10.1038/s41467-024-54606-2.

[10] Q. Sun et al., "Recent developments in the PySCF program package," *Journal of Chemical Physics* **153**, 024109 (2020). https://doi.org/10.1063/5.0006074.

[11] M. Onizhuk and G. Galli, "PyCCE: A Python Package for Cluster Correlation Expansion Simulations of Spin Qubit Dynamics," *Advanced Theory and Simulations* **4**, 2100254 (2021). arXiv:2107.05843; doi:10.1002/adts.202100254.

[12] N. Lambert et al., "QuTiP 5: The Quantum Toolbox in Python," *Physics Reports* **1153**, 1–62 (2025). arXiv:2412.04705.

[13] T. P. Ogden, *MaxwellBloch*, software and official documentation, version used: 0.12.0. https://github.com/tpogden/maxwellbloch; https://maxwellbloch.readthedocs.io/.

[14] C. E. Bradley et al., "Robust quantum-network memory based on spin qubits in isotopically engineered diamond," *npj Quantum Information* **8**, 122 (2022). https://doi.org/10.1038/s41534-022-00637-w.

[15] C. E. Bradley, "Data/software underlying the publication: Robust quantum-network memory based on spin qubits in isotopically engineered diamond," 4TU.ResearchData, version 3 (2022), CC BY 4.0. https://doi.org/10.4121/16887658.v3.

[16] S. DiAdamo, J. Nötzel, B. Zanger and M. M. Beşe, "QuNetSim: A Software Framework for Quantum Networks," *IEEE Transactions on Quantum Engineering* **2**, 1–12 (2021). https://doi.org/10.1109/TQE.2021.3092395.

[17] X. Wu, A. Kolar, J. Chung, D. Jin, T. Zhong, R. Kettimuthu and M. Suchara, "SeQUeNCe: a customizable discrete-event simulator of quantum networks," *Quantum Science and Technology* **6**, 045027 (2021). https://doi.org/10.1088/2058-9565/ac22f6.

[18] S. P. Huber et al., "AiiDA 1.0, a scalable computational infrastructure for automated reproducible workflows and data provenance," *Scientific Data* **7**, 300 (2020). https://doi.org/10.1038/s41597-020-00638-4.

[19] J. Chung et al., "Cross-Validation of Open-Source Quantum Network Simulators," arXiv:2610.09322v1 (2026). https://arxiv.org/abs/2610.09322.